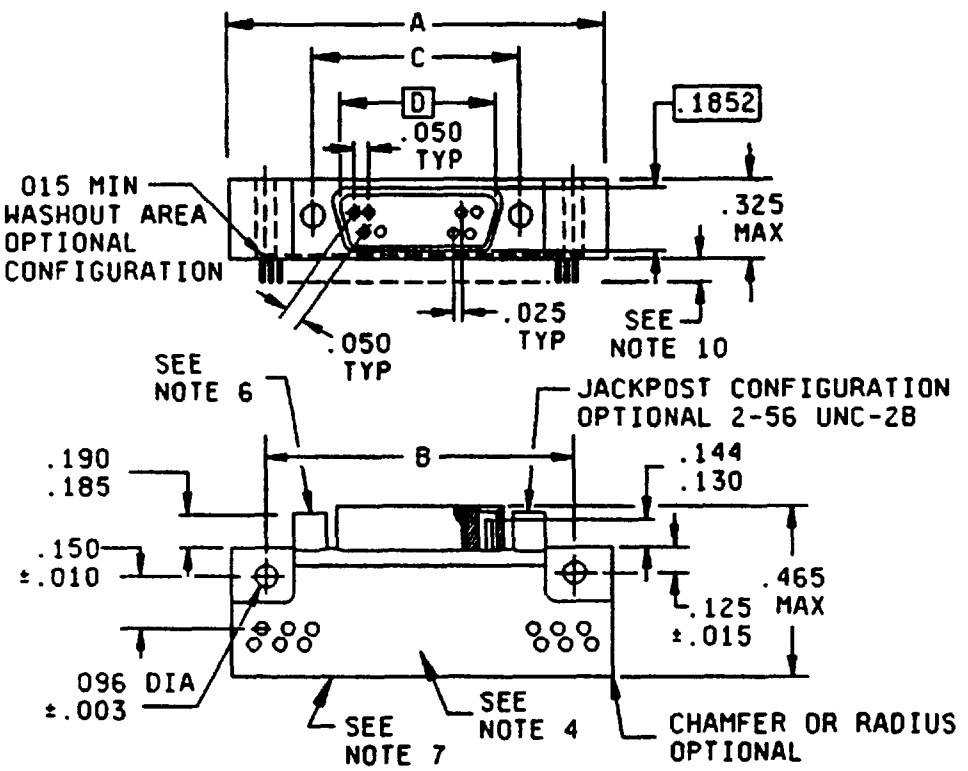
INCH-POUND
MIL-C-83513/19A
14 Oct 1993
SUPERSEDING
MIL-C-83513/19
27 February 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, SOCKET CONTACTS, 2 ROW ,SOLDER TYPE, STANDARD PROFILE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513.

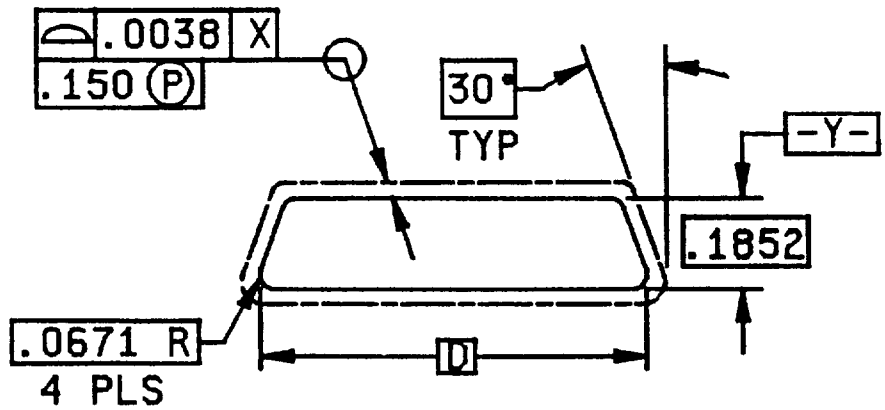


Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.015	0.38	.0671	1.704	.144	3.66	.190	4.83
.0038	0.097	025	0.64	.096	2 44	.150	3 81	.200	5 08
.005	0 13	.050	1.27	.125	3 18	.185	4.70	.325	8.26
010	0 25	05	1 3	.130	3.30	.1852	4.704	.465	11.81

FIGURE 1 Connector, receptacle, .050 spacing.

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.



DETAIL OF INTERFACE

Number of contacts	A max	B ±.007 (0.18)	C ±.005 (0.13)	D min
9	1.390 (35.31)	1.150 (29.21)	.565 (14.35)	.3342 (8 489)
15	1.540 (39.12)	1.30 (33.02)	.715 (18.16)	.4842 (12.299)
21	1.690 (40.64)	1 450 (36.83)	.865 (21.97)	.6342 (16.109)
25	1.790 (45.47)	1.550 (39 37)	.965 (24.51)	.7342 (18.649)
31	2.040 (51.82)	1.80 (45.7)	1.115 (28.32)	.8842 (22 459)
37	2.340 (59.44)	2.10 (53.3)	1.265 (32.13)	1.0342 (26.269)

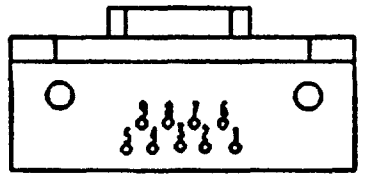
NOTE. Metric equivalents are in parentheses.

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0 13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513. Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
5. Metal shell shall be of material in accordance with MIL-C-83513 for class M parts
6. Jackpost (permanently attached), when specified: Corrosion-resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.
7. Separately molded plastic body (if used) shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519
8. Wire termination sockets shall conform to QQ-W-343, no. 24 AWG copper.
9. Interfacial seals are included with each type receptacle connector
10. Termination lengths available .109 (2.77 mm), 140 (3.56 mm), or .172 (4.37 mm) The tolerance shall be ±.015 (0.38 mm) for all termination lengths.

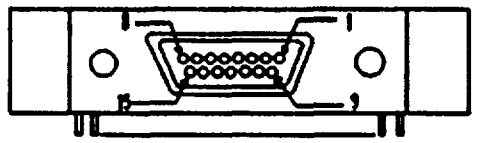
FIGURE 1. Connector, receptacle, 050 spacing - Continued

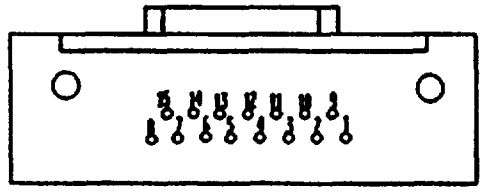
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.



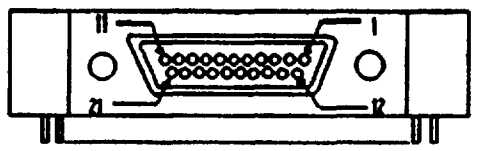


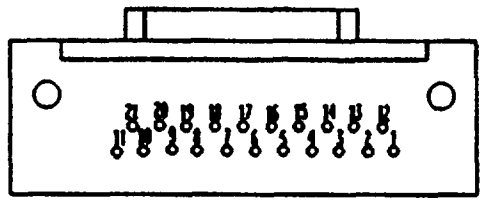
9 CAVITY





15 CAVITY



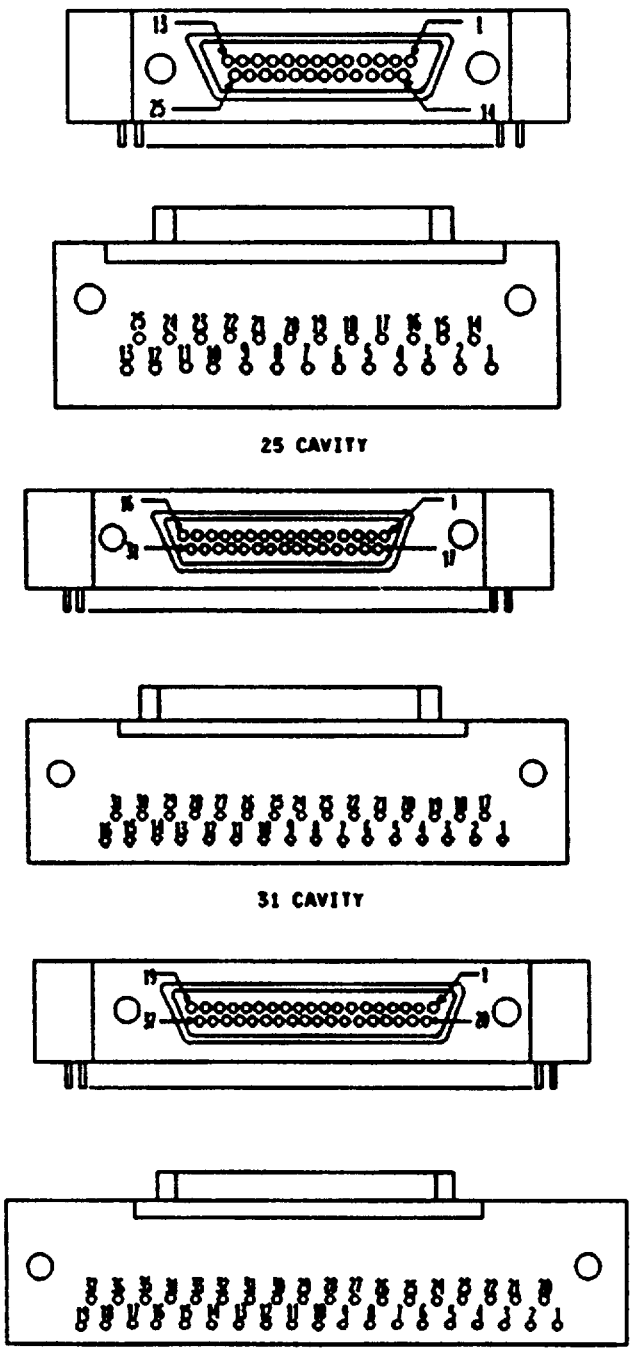


21 CAVITY

NOTE: Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part

FIGURE 2 Insert arrangement

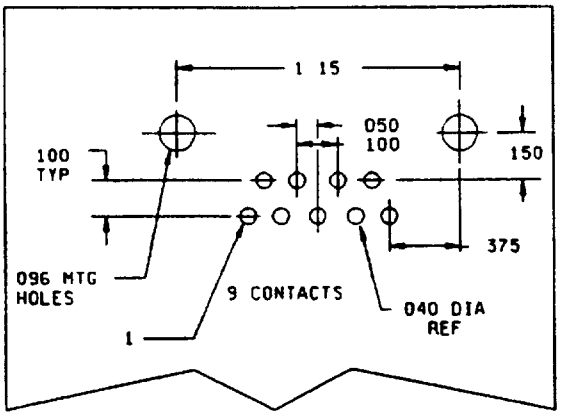
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

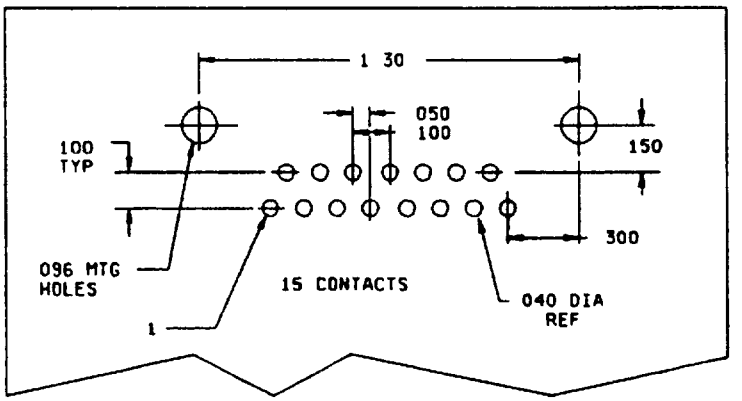


NOTE: Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2 Insert arrangement - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.





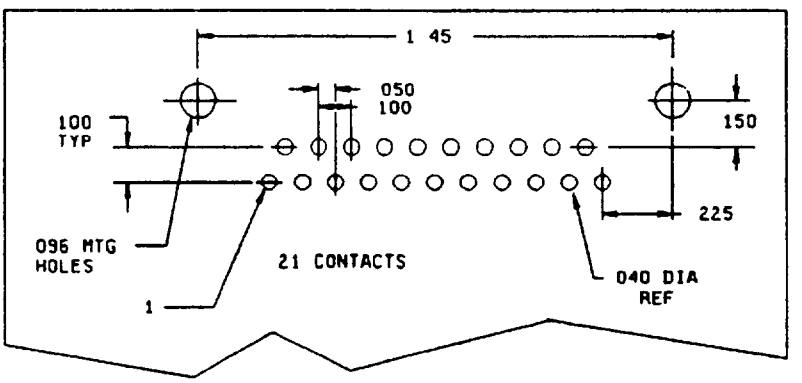
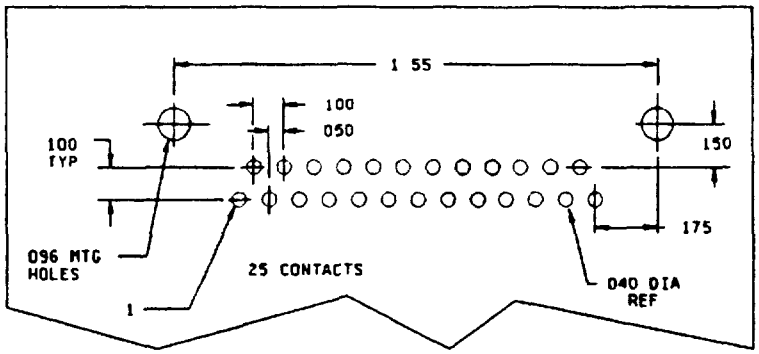
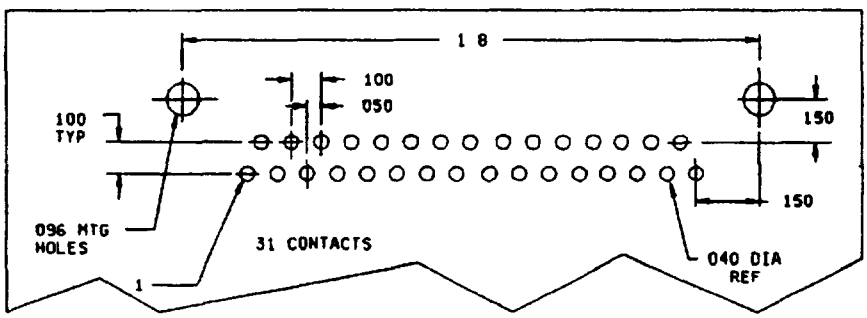
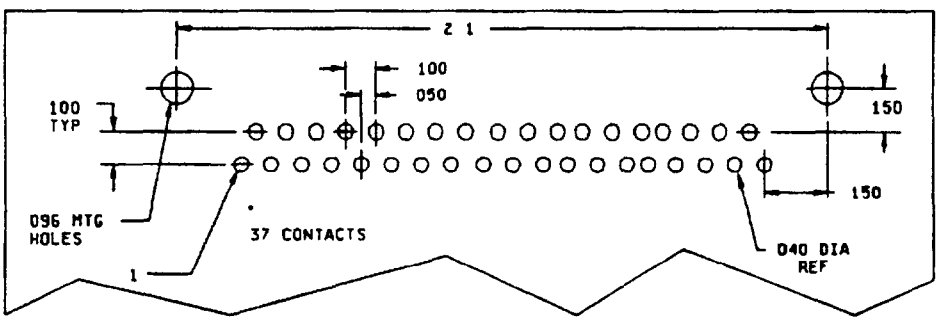


FIGURE 3. Layout arrangement

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.







Inches	mm	Inches	mm	Inches	mm
.040	1.02	.175	4.44	1.30	33.02
050	1.27	.225	5 72	1.45	36.83
096	2.44	.300	7.62	1 55	39 37
.100	2.54	.375	9.53	1 8	45.7
.150	3.81	1.15	29.21	2.1	53.3

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for information only
3. Pattern shown is for connector mounting side of PC board

FIGURE 3. Layout arrangement - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations See figures 1 and 2

Current rating, maximum 3 amperes per contact

Materials

Termination organization area: Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513.

Shell: The requirements for shell materials shall be in accordance with MIL-C-83513.

Plastic body or plastic molding: Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.

Jackpost: Corrosion-resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.

Wire termination pins. Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.

Mating connector: Shall conform to MIL-C-83513/1 and MIL-C-83513/3.

Plating of termination leads: Solder dipping socket of termination leads will be accomplished in Sn60 Pb40 or Sn63 Pb37 in accordance with QQ-S-571.

Part or Identifying Number (PIN): Consists of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code for the shell finish and the hardware.

M83513/19-	D	01	C	P
Specification sheet number	Insert arrangements (see figure 2) A = 09 B = 15 C = 21 D = 25 E = 31 F = 37	Termination length 01 = .109 02 = 140 03 = .172	Shell finish C = Cadmium N = Electroless nickel (space application)	Hardware N = No jackpost P = Jackpost attached

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS, SH
Air Force - 17, 99
DLA - ES

User activities
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent
DLA - ES

(Project 5935-3921-10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:29Z
Check the source to verify that this is the current version before use.